Vancomycin Dosing and Monitoring
Version 1.2

Purpose: To guide physicians and pharmacists on different dosing and monitoring regimens of injectable vancomycin, using evidence-based guidelines and best practices.

Policy: Applicable to all pharmacists and clinicians

Guideline: Intravenous Vancomycin Dosing & Monitoring Regimen in Adults

Introduction: Therapeutic Drug Monitoring

Most PK/PD data generated on vancomycin has focused on treatment of serious methicillin-resistant Staphylococcus aureus (MRSA) infections. Therefore, extrapolation of these recommendations to methicillin-susceptible strains, coagulase-negative staphylococci, and other pathogens should be viewed with extreme caution.

Indication for trough level monitoring

Vancomycin monitoring is recommended for

- Patients receiving high dosing for MRSA infections to achieve sustained targeted AUC.
- All patients at high risk of nephrotoxicity (e.g., critically-ill patients receiving concurrent nephrotoxins).
- Patients with unstable (i.e., deteriorating or significantly improving) renal function
- Patients receiving prolonged courses of therapy (more than three to five days).

Vancomycin dose

Actual body weight to be used to determine the vancomycin loading and maintenance doses for severe infections in patients who were seriously ill.
Vancomycin dosages of 15–20 mg/kg (based on as actual body weight) given every 8–12 hours are required for most patients with normal renal function to achieve the suggested serum concentrations when the MIC is ≤1 mg/L. With this vancomycin dosing regimen, AUC/MIC ratio of 400-600 is required for best clinical effectiveness and less likelihood of nephrotoxicity.

A targeted AUC/MIC of ≥400 is not achievable with conventional dosing methods if the vancomycin MIC is ≥2 mg/L in a patient with normal renal function (i.e., Crcl of 70–100 mL/min). Therefore, alternative therapies should be considered.

Initial Loading Dose

- Loading doses are recommended in serious MRSA infections such as sepsis, meningitis, bacteremia, infective endocarditis, pneumonia, and osteomyelitis, patients who are critically-ill or in the intensive care unit, requiring dialysis or renal replacement therapy, or receiving continuous infusion therapy of vancomycin.
- Initial loading dose of **25– 30 mg/kg** for both intermittent and continuous infusion administration of vancomycin (based on actual body weight and **not exceed 3000 mg**) can be used to facilitate rapid attainment of target trough serum vancomycin.

Standard load for patients with normal renal function: 25-30mg/kg TBW

Patient Weight	Standard Loading Dose ~25 mg/kg TBW	Modified Loading Dose 15-20 mg/kg TBW Obese (BMI ≥ 30) CrCL < 30 or AKI, IHD, CRRT, unavailable Scr in emergent situations (e.g code sepsis or ED)
36 – 45 kg	1,000 mg x 1	750 mg x 1
46 – 55 kg	1,250 mg x 1	1,000 mg x 1
56 – 65 kg	1,500 mg x 1	1,250 mg x 1
66 – 75 kg	1,750 mg x1	1,500 mg x 1
76 – 120 kg	2,000 mg x 1	1,750 mg x1
> 120 kg	2,000 mg x 1	2,000 mg x 1

*Time maintenance dose start based on renal function: e.g. wait 24h to start maintenance regimen if CrCl = 30
Use total body weight (TBW); Round doses to nearest 250mg. Infuse each 1000mg over 60 minutes.

- Trough samples should be obtained within **30 minutes before** the 3rd or 4th dose (before 3rd dose if once daily) in patients with normal renal function.
- Routine levels should be ordered no more than twice weekly if stable and within target range.
- Levels should be checked daily when clinically warranted i.e. in patients with acute renal impairment, CrCl <10 mg/ min, concurrent use of nephrotoxic medications, high risk of nephrotoxicity or haemodynamically unstable.
- In patients on haemodialysis (HD), levels should be checked pre each HD session.
- Follow up levels subsequent to a dose adjustment should be ordered 36-48 hours after the new dosing regimen has started.
- **When plasma levels are unexpectedly high or low, confirm the timing of sampling and the administration history/record of vancomycin doses with the relevant nursing staff. Such levels may reflect missed doses or sampling performed during or shortly after vancomycin infusions rather than true trough levels.**

Maintenance Dosing

Doses of 15–20 mg/kg (ABW) given every 8–12 hr for most patients with normal renal function to achieve the suggested serum concentrations when the MIC is ≤1 mg/L (Maximum maintenance dose is 2 g per dose and 4.5 g in the first 24 hrs).

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Creatinine Clearance (mL/min)	Dose & Frequency Total body weight (TBW)	TDD Range	Timing of Peak/Trough Levels
> 90	15 mg/kg Q8-12H BMI ≥ 30: 10 – 15 mg/kg TBW Q12H† BMI ≥ 40: 7.5 – 12.5 mg/kg TBW Q12H†	30 – 45 mg/kg/day Obese: 15 – 30 $mg/kg_{TBW}/day$	Peak 1hr after 4th / trough 30 min before 5th dose, or Peak 1hr after 3rd/ trough 30 min before 4th dose
51-89	10– 20 mg/kg Q12H BMI ≥ 30: 10 – 12.5 mg/kg TBW Q12H† BMI ≥ 40: 7.5 – 10 mg/kg TBW Q12H†	20– 40 mg/kg/day Obese: 15 – 25 $mg/kg_{TBW}/day$	Q12H: Peak 1hr after 4th / trough 30 min before 5th dose, or Peak 1hr after 3rd/ trough 30 min before 4th dose
30-50	10-15 mg/kg Q12H to 20 mg/kg Q24H	20 – 30 mg/kg/day	Q12H: as above Q24H: Peak 1hr after 3rd/ trough 30 min before 4th dose
10-29	10 – 15 mg/kg Q24H to 15 mg/kg Q48H	7.5 – 15 mg/kg/day	Q24H – Peak 1hr after 3rd/ trough 30 min before 4th dose Q48H – Peak 1hr after 2nd dose; trough 30 min before 3rd dose
<10 or AKI*, dose by level	15 mg/kg x1, then dose by level	N/A	Trough within 24 hours of last dose, or with AM labs or every other day

Obese patients

Vancomycin loading dose of 20-25 mg/kg using actual body weight with a maximum of 3000 mg may be considered in obese adult patients with serious infections. Empiric maintenance doses ≤4500 mg/day are expected for most obese patients.
For obese patients, initial dosing can be based on ABW and then adjusted based on serum vancomycin concentrations to achieve therapeutic levels. More intensive therapeutic monitoring should also be performed in obese patients. Measurement of peak and trough concentrations is recommended to improve the accuracy of vancomycin AUC estimation and maintenance dose optimization in obese patients.

Renal replacement therapy

Creatinine Clearance (mL/min)	Dose & Frequency Total body weight (TBW)	TDD Range	Timing of Peak/Trough Levels
Hemodialysis	Initial: 15 – 20 mg/kg x 1 (max 2gm) Maintenance: see appendix E	N/A	• Single pre-dialysis level (preferred) • Alternative: single level 4 hours after completion of dialysis session
CRRT‡	Initial: 15 – 20 mg/kg x 1 (max 2gm) Maintenance: 10 – 15 mg/kg Q24H	N/A	Trough 30 min before 3rd or 4th dose
Peritoneal dialysis	10 – 15 mg/kg IV x1, then dose by level	N/A	Check level 24h after initial dose. Consult ASP
	Dosing for intraperitoneal (IP) instillation (*NOT part of protocol*) [Li, 2016] Intermittent (1 exchange/day): 15-30mg/kg IP initially, then dose by level* *supplemental doses may be needed for APD patients		Intraperitoneal dosing (*off-protocol*): Level with AM labs on day 3 after any dose administered (allow fluid redistribution before drawing random level)

† Note: For those with $CrCL_{adjBW}$ > 120mL/min, Q8H may be considered if t½ < 8hr**
‡ *Loading and maintenance doses are based on 1-2L/hr dialysate flow and ultrafiltration rates, which is estimated to mimic a creatinine clearance of 30-50 mL/min*

Dose adjustment in patients on haemodialysis (HD), based on pre-HD level

Pre HD vancomycin level	Intervention
< 10 mg/mL	Give 750-1000 mg
10-15 mg/mL	Give 500-750 mg
15-20 mg/mL	Give 250-500 mg
20-25 mg/mL	Hold 1 dose OR give 250 mg
> 25 mg/mL	Hold vancomycin until level back in range (check level 4-6 hrs after next HD session, re-dose if level < 20-25 mg/mL)

Monitoring frequency of vancomycin serum level

- Trough samples should be obtained within **one hour before** the 3rd or 4th dose (before 3rd dose if once daily) in patients with normal renal function.
- Routine levels should be ordered no more than twice weekly if stable and within target range.
- Levels should be checked daily when clinically warranted i.e. in patients with acute renal impairment, Crcl < 10 mg/ min, concurrent use of nephrotoxic medications, high risk of nephrotoxicity or haemodynamically unstable.
- In patients on haemodialysis (HD), levels should be checked pre each HD session.
- Follow up levels subsequent to a dose adjustment should be ordered 36-48 hours after the new dosing regimen has started.

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Vancomycin Continuous infusion

- The pharmacokinetics of continuous infusions suggest that such regimens may be a reasonable alternative to conventional dosing and provide a convenient way to readily achieve the desired vancomycin therapeutic range (i.e., steady-state concentration of 20 – 25 mg/L) throughout the entire dosing period.
- The risk of developing nephrotoxicity with continuous infusion appears to be similar or lower compared to intermittent dosing when targeting steady-state concentration 15-25mg/L and trough 10-20 mg/L respectively.

TDM for Vancomycin-Induced Nephrotoxicity

- SCr change by ≥ 0.3 mg/dL within 48h or
- SCr increase by 50% from baseline in consecutive daily reading **OR**
- CrCl change by >25 - 50% from baseline on two consecutive day **OR**
- Urine output < 0.5 mL/kg/hr over 6 hours (oliguria)

Concurrent administration of nephrotoxic agents such as aminoglycosides, loop diuretics, amphotericin B, and vasopressors and piperacillin-tazobactam increase the risk of nephrotoxicity.

Vancomycin-Induced Ototoxicity

Monitoring for high frequency auditory loss should be considered for patients receiving additional ototoxic agents, such as aminoglycosides.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
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Review / Revision History

	Revision History		
Revision	**Description of change**	**Author**	**Revision Date**
1	Initial Release	Microbiology	3/6/2022
2	Revision	Clinical Pharmacy	21/01/2023
3			
4			
5			

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